

Day 4 Problem 3

SUBMISSION BY :

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BUG 02 · db_queries.py 2 bugs

Querying Attendance Records from the Database

The function below queries attendance records for a specific employee in a given month. It is returning incorrect data and in some cases returning records for all employees.

```
def get_monthly_attendance(conn, employee_id, month, year):
    query = """
        SELECT a.date, a.check_in, a.check_out, e.name
        FROM attendance a
        LEFT JOIN employees e      ← Bug A
        ON a.employee_id = e.id
        WHERE a.month = ?         ← Bug B
        AND a.year = ?
        ORDER BY a.date ASC
    """

    cursor = conn.cursor()
    cursor.execute(query, (month, year))
    return cursor.fetchall()
```

Symptoms

A — Records are returned even when there is no matching employee in the `employees` table. Only attendance records with a valid, existing employee should be returned.

B — The query is returning attendance records for all employees in that month, not just the specified one. The `employee_id` filter is never applied.

Your tasks

1. Change the JOIN type so only records with a matching employee are returned.
2. Add the missing `employee_id` filter to the WHERE clause and pass it correctly to the query.

SQL bug ~10 min

Solution:

BUG A-

LEFT JOIN employees e

Records are returned even when there is no matching employee in the `employees` table. Only attendance records with a valid, existing employee should be returned.

Fix

INNER JOIN employees e

It ensures only employees present with a valid employee should be returned.

BUG B-

WHERE a.month = ?

AND a.year = ?

Employee_id filter not applied

Fix:

WHERE a.employee_id = ?

AND a.month = ?

AND a.year = ?

BUG 03 · attendance.py 2 bugs

Calculating Working Hours for the Day

This function calculates the number of hours an employee worked in a day and flags them as late if they checked in after 9:00 AM. Both calculations are producing wrong results.

```
from datetime import datetime

WORK_START = "09:00"
LATE_THRESHOLD_MINUTES = 15

def calculate_hours(check_in: str, check_out: str):
    fmt = "%H:%M"
    ci = datetime.strptime(check_in, fmt)
    co = datetime.strptime(check_out, fmt)

    duration = ci - co  # ← Bug A
    hours_worked = duration.seconds / 3600

    start = datetime.strptime(WORK_START, fmt)
    late_by = (ci - start).seconds // 60

    is_late = late_by > LATE_THRESHOLD_MINUTES  # ← Bug B

    return round(hours_worked, 2), is_late
```

Symptoms

A — hours_worked is always negative or producing a very large incorrect number. The subtraction operands are in the wrong order.

B — An employee who checks in exactly 15 minutes late is not being flagged, but they should be. The boundary condition is off by one.

Your tasks

1. Fix the subtraction so duration represents time from check-in to check-out.
2. Correct the comparison operator so an employee who is exactly 15 minutes late is also flagged.

Logic bug Off-by-one ~12 min

Solution:

BUG A-

duration = ci – co

Fix:

duration = co – ci

BUG B-

is_late = late_by > LATE_THRESHOLD_MINUTES

Fix

is_late = late_by >= LATE_THRESHOLD_MINUTES

So that the employee who comes late exactly by the threshold should also be marked late

BUG 04 · report.py 2 bugs

Generating the Monthly Attendance Summary

This function takes a list of daily attendance records and produces a summary. It is crashing on some inputs and calculating the average hours incorrectly.

```
def generate_summary(records):
    total_hours = 0
    late_days = 0

    for i in range(len(records) + 1):  ← Bug A
        record = records[i]
        total_hours += record["hours"]
        if record["is_late"]:
            late_days += 1

    avg_hours = total_hours / len(records)  ← Bug B

    return {
        "total_hours": round(total_hours, 2),
        "avg_hours": round(avg_hours, 2),
        "late_days": late_days,
        "days_present": len(records)
    }
```

Symptoms

A — The function raises an `IndexError` on every call. The loop iterates one step too far beyond the list length.

B — When an employee has zero attendance records, the function crashes with a `ZeroDivisionError` instead of returning `0` for average hours.

Your tasks

1. Fix the loop range so it does not go out of bounds.
2. Handle the empty records case — return `avg_hours: 0` instead of dividing by zero.

Off-by-one Runtime error ~12 min

Solution:

BUG A-

for i in range(len(records) + 1):

Fix

for i in range(len(records)):

Accessing index within length of records not exceeding it.

BUG B-

avg_hours = total_hours / len(records)

Fix

if len(records) == 0:

avg_hours = 0

else:

avg_hours = total_hours / len(records)

When there are no attendance records, dividing by zero causes a crash. The average should be 0.

BUG 05 · report.py 2 bugs

Flagging Employees Below Minimum Attendance

HR policy requires that employees must be present for at least 20 days per month and must not exceed 3 late check-ins before a warning is issued. Both conditions in this function have logic errors.

```
MIN_DAYS_REQUIRED = 20
MAX_LATE_DAYS = 3

def check_attendance_policy(summary):
    days_present = summary["days_present"]
    late_days = summary["late_days"]

    below_minimum = days_present > MIN_DAYS_REQUIRED  ← Bug A
    exceeded_late = late_days > MAX_LATE_DAYS          ← Bug B

    if below_minimum or exceeded_late:
        return {
            "warning": True,
            "reason": []
            + (["Below minimum attendance"] if below_minimum else [])
            + (["Exceeded late check-ins"] if exceeded_late else [])
        }
    return {"warning": False, "reason": []}
```

Symptoms

A — Employees who attend fewer than 20 days are *not* being flagged. Employees who attend more than 20 days are being flagged instead. The comparison is inverted.

B — An employee with exactly 3 late check-ins is not receiving a warning, but HR policy states that 3 or more late check-ins should trigger one.

Your tasks

1. Fix the `below_minimum` condition so it correctly flags employees who attended fewer than the minimum required days.
2. Fix the `exceeded_late` condition so employees with exactly `MAX_LATE_DAYS` late check-ins are also flagged.

Logic bug Off-by-one ~12 min

Solution:

BUG A-

```
below_minimum = days_present > MIN_DAYS_REQUIRED
```

Fix:

```
below_minimum = days_present < MIN_DAYS_REQUIRED
```

Employees should be flagged when they attend **fewer than 20 days**, not more than 20 days.

BUG B-

```
exceeded_late = late_days > MAX_LATE_DAYS
```

Fix

```
exceeded_late = late_days >= MAX_LATE_DAYS
```

HR policy states that **3 or more** late check-ins should trigger a warning, so `>=` is required.